

Polysomnography (PSG), Sleep Center

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Link to Codes

MCG Health
Ambulatory Care
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Clinical Indications

- Sleep center PSG may be indicated for **1 or more** of the following(1)(2):
 - ☐ Adult with obesity hypoventilation syndrome, suspected, as indicated by **ALL** of the following(17)(18)(19):
 - BMI greater than 30  BMI Calculator
 - Daytime hypercapnia with PaCO₂ greater than 45 mm Hg (6.0 kPa) without other etiology (eg, kyphoscoliosis, lung parenchymal disease, myopathy, severe hypothyroidism)
 - TSH level does not demonstrate hypothyroidism.(22)
 - ☐ Adult with obstructive sleep apnea, suspected, as indicated by **1 or more** of the following(8)(9)(23)(24)(25):
 - Initial sleep center PSG test, as indicated by **ALL** of the following:
 - Home PSG is not appropriate, as indicated by **1 or more** of the following(33)(34):
 - Chronic opioid use
 - Clinically significant neurologic or neuromuscular disease (eg, amyotrophic lateral sclerosis, Charcot-Marie-Tooth disease, Duchenne muscular dystrophy)(35)(36)(37)(38)
 - Complex sleep disorder, suspected (eg, narcolepsy, cataplexy, periodic limb movement disorder)
 - Home PSG services not available
 - Hypoventilation syndrome[A](39)
 - Moderate-to-severe chronic obstructive pulmonary disease or other lung disease (eg, FEV₁ less than 60% predicted)[B](10)(40)
 - Moderate-to-severe heart failure (eg, New York Heart Association functional class III or IV)[C](10)(35)
 - Nocturnal oxygen use
 - Ongoing suspicion of obstructive sleep apnea despite initial negative home sleep study[D](42)(43)
 - Patient unable to properly operate or tolerate home study equipment
 - Previous home PSG results inadequate for suspected obstructive sleep apnea or upper airway resistance syndrome[E](44)
 - Sleep center PSG coincident with CPAP titration as split-night study (ie, performed if moderate or severe obstructive sleep apnea documented during first 2 hours, minimum, of study night and at least 3 hours remain in the night for the CPAP titration component)[F](27)(30)
 - Signs or symptoms suggestive of moderate-risk to high-risk obstructive sleep apnea, including **1 or more** of the following:

- Epworth Sleepiness Scale score of 10 or greater[G](3)(4)
- Excessive daytime sleepiness, fatigue, or awakening with gasping or choking, and high risk for injury (eg, falling asleep while driving)(45)(46)
- Excessive daytime sleepiness, fatigue, or awakening with gasping or choking, and risk factor for sleep apnea, as indicated by **1 or more** of the following(5)(47):
 - BMI greater than 30  BMI Calculator
 - Resistant hypertension,[H] as indicated by **ALL** of the following(48)(49)(50)(51):
 - Medication regimen includes 3 or more antihypertensive drugs at therapeutic dosages, including one diuretic.
 - Office blood pressure above goal
- Observed apnea or choking episodes
- Significant oxygen desaturation[I] (eg, less than 90%) on overnight pulse oximetry(4)
- Snoring
- Repeat sleep center PSG test, as indicated by **ALL** of the following(29)(31)(32)(55)(56):
 - Appropriate testing situation, as indicated by **1 or more** of the following:
 - Assessment of treatment adequacy independent of symptoms (eg, CPAP device-generated data of concern, first-episode transient ischemic attack, new-onset atrial fibrillation)[J]
 - Bariatric surgery performed in adult with known obstructive sleep apnea(56)
 - Oral appliance to treat obstructive sleep apnea after final adjustments of fit(57)(58)
 - Postintervention signs or symptoms suggestive of persistent moderate-risk to high-risk obstructive sleep apnea, including **1 or more** of the following:
 - Epworth Sleepiness Scale score of 10 or greater[G](3)(4)
 - Excessive daytime sleepiness, fatigue, or awakening with gasping or choking, and high risk for injury (eg, falling asleep while driving)(45)(46)
 - Excessive daytime sleepiness, fatigue, or awakening with gasping or choking, and risk factor for sleep apnea, as indicated by **1 or more** of the following(5)(47):
 - BMI greater than 30  BMI Calculator
 - Resistant hypertension,[H] as indicated by **ALL** of the following(48)(49)(50)(51):
 - Medication regimen includes 3 or more antihypertensive drugs at therapeutic dosages, including one diuretic.
 - Office blood pressure above goal
 - Observed apnea or choking episodes
 - Significant oxygen desaturation[I] (eg, less than 90%) on overnight pulse oximetry(4)
 - Snoring
 - Postoperative assessment of efficacy of surgery to treat obstructive sleep apnea (eg, upper airway procedures, tracheostomy)(55)
 - Suspicion of obstructive sleep apnea due to new signs or symptoms (eg, weight gain accompanied by symptoms, new nocturia) in patient with previous negative study(59)(60)
 - Weight loss greater than 10% in patient using CPAP to determine if CPAP can be discontinued
 - Home PSG is not appropriate, as indicated by **1 or more** of the following(33):
 - Clinically significant neurologic or neuromuscular disease (eg, amyotrophic lateral sclerosis, Charcot-Marie-Tooth disease, Duchenne muscular dystrophy)(35)(36)(37)(38)
 - Complex sleep disorder, suspected (eg, narcolepsy, cataplexy, periodic limb movement disorder)
 - Home PSG services not available
 - Hypoventilation syndrome[A]
 - Moderate-to-severe chronic obstructive pulmonary disease or other lung disease (eg, FEV₁ less than 60% predicted)[B](10)(40)
 - Moderate-to-severe heart failure (eg, New York Heart Association functional class III or IV)[C](10)(35)
 - Nocturnal oxygen use
 - Ongoing suspicion of obstructive sleep apnea despite initial negative home sleep study[D](42)(43)
 - Patient unable to properly operate or tolerate home study equipment
 - Previous home PSG results inadequate for suspected obstructive sleep apnea or upper airway resistance syndrome[E](44)
 - Sleep center PSG coincident with CPAP titration as split-night study (ie, performed if moderate or severe obstructive sleep apnea documented during first 2 hours, minimum, of study night and at least 3 hours remain in the night for the CPAP titration component)[F](27)(30)

-  Central sleep apnea or sleep-related hypoventilation, suspected, as indicated by **ALL** of the following(57)(61)(62)(63):[N](#)
- Appropriate clinical condition, as indicated by **1 or more** of the following:
 - Brainstem lesion(70)
 - Chiari malformation(68)(69)(70)(71)(72)
 - Congenital central alveolar hypoventilation syndrome

- Down syndrome(71)(73)(74)
- Heart failure
- Neuromuscular or chest wall disorder (eg, amyotrophic lateral sclerosis, Duchenne muscular dystrophy, kyphoscoliosis, myotonic dystrophy, spinal cord injury)(36)(75)(76)
- Prader-Willi syndrome(67)
- Skeletal dysplasia (eg, achondroplasia)(71)(77)(78)
- Signs or symptoms suggestive of central sleep apnea or sleep-related hypoventilation, as indicated by **1 or more** of the following:
 - Difficulty falling asleep or remaining asleep
 - Dyspnea on awakening
 - Excessive daytime sleepiness
 - Nonrestorative sleep
 - Unexplained clinical findings suggestive of hypoventilation (eg, cor pulmonale, cyanosis, low oxygen saturation when breathing room air, shallow respirations)(66)
 - Unexplained laboratory abnormalities suggestive of hypoventilation (eg, elevated arterial carbon dioxide during spontaneous breathing, erythrocytosis)(66)

☐ Child, infant, or adolescent with obstructive sleep apnea, suspected, and **1 or more** of the following(12)(57)(63)(85)(86):^[I]

- Initial PSG test, as indicated by **ALL** of the following(101)(102):
 - Appropriate clinical situation, as indicated by **1 or more** of the following(91):
 - Adenoid or tonsillar enlargement, and adenotonsillectomy being considered for treatment
 - Chiari malformation(68)(72)(103)
 - Craniofacial malformation (eg, Apert syndrome, Crouzon syndrome, Pierre Robin sequence)(104)(105)(106)
 - Down syndrome(107)(108)(109)(110)(111)
 - Neuromuscular disorder (eg, Duchenne muscular dystrophy, myotonic muscular dystrophy)(36)(112)
 - Signs and symptoms consistent with obstructive sleep apnea, including **1 or more** of the following:
 - Daytime sleepiness
 - Enuresis
 - Failure to thrive (weight less than fifth percentile for age)(100)
 - Hyponasal speech
 - Mouth breathing
 - Nocturnal pauses in breathing
 - Nonspecific behavioral problems (eg, hyperactivity, developmental delay, aggression, poor school performance)
 - Pulmonary hypertension
 - Signs of increased respiratory effort (ie, nasal flaring)
 - Snoring(100)
 - Skeletal dysplasia (eg, achondroplasia)(77)(78)(113)(114)
 - Appropriate test requested, as indicated by **1 or more** of the following:
 - Full-night PSG
 - Split-night study (ie, PSG with CPAP titration) for patient age 12 years or older (ie, performed if moderate or severe obstructive sleep apnea documented during first 2 hours, minimum, of study night and at least 3 hours remain in the night for the CPAP titration component)(M)(29)(115)
- Repeat PSG test, as indicated by **ALL** of the following:
 - Appropriate clinical situation, as indicated by **1 or more** of the following:
 - Assessment of treatment adequacy independent of symptoms (eg, CPAP device-generated data of concern, new systemic hypertension, unexpected slowing of growth rate)^[N]
 - Clinical assessment suggests obstructive sleep apnea in patient with prior negative PSG testing.
 - Growing child, and assessment of ongoing need for pressure therapy(117)
 - Need to perform postoperative assessment in patient at risk of inadequate response to intervention (eg, post adenotonsillectomy), as indicated by **1 or more** of the following(57)(91)(99):
 - Apnea-hypopnea index or respiratory disturbance index 20 or greater on preoperative PSG
 - BMI greater than 95th percentile for age(118)
 - Craniofacial anomalies that obstruct upper airway
 - Neurologic disorder (eg, Down syndrome, Prader-Willi syndrome, myelomeningocele)(37)(67)
 - Ongoing signs and symptoms suggestive of persistent obstructive sleep apnea
 - Oral appliance treatment of obstructive sleep apnea, and need to document efficacy of therapy
 - Rapid maxillary expansion, and need to document efficacy of therapy
 - Appropriate test requested, as indicated by **1 or more** of the following:
 - Full-night PSG
 - Split-night study (ie, PSG with CPAP titration) for patient age 12 years or older (ie, performed if moderate or severe obstructive sleep apnea documented during first 2 hours, minimum, of study night

and at least 3 hours remain in the night for the CPAP titration component)(M)(29)(115)

- ☐ Insomnia^[O] and **1 or more** of the following(119):^[N]
 - Inadequate response to treatment of insomnia by neurologist or sleep specialist
 - Periodic limb movement disorder, suspected
 - Precipitous arousals with injurious behavior, suspected
 - Sleep-related breathing disorder, suspected, as indicated by **1 or more** of the following^[P]:
 - Excessive daytime sleepiness
 - Observed apnea or choking episodes
 - Snoring
- ☐ Narcolepsy, suspected, as indicated by **ALL** of the following(121)(122)(123):^[N]
 - Signs or symptoms suggestive of narcolepsy, as indicated by **1 or more** of the following:
 - Cataplexy (ie, episode of sudden bilateral loss of postural tone associated with intense emotion)
 - Excessive daytime sleepiness
 - Hallucinations with sleep onset (hypnagogic) or upon awakening (hypnopompic)
 - Ongoing suspicion of narcolepsy after negative initial study, or appearance of new symptoms contributing to sleepiness(126)(127)(128)
 - Recurrent daytime naps or lapses into sleep for over 3-month period
 - Sleep paralysis (ie, brief episodic loss of voluntary movement that occurs during period of falling asleep or when awakening)
 - PSG precedes multiple sleep latency test. See Multiple Sleep Latency Test (MSLT) and Maintenance of Wakefulness Test (MWT) ^[AC] for further information.
- ☐ Parasomnia,^[Q] as indicated by **1 or more** of the following(129):^[N]
 - Confusional arousal
 - Inadequate response to treatment
 - New onset in adult
 - Rapid eye movement sleep behavior disorder, suspected(123)(133)
 - Sleep-related eating disorder
 - Sleep terrors, and need to rule out other diagnosis (eg, comorbid sleep-disordered breathing, nocturnal seizure, rapid eye movement sleep behavior disorder)(133)
- ☐ Periodic limb movement disorder^[R] and **1 or more** of the following:^[N]
 - Excessive daytime sleepiness
 - Observed apnea or choking episodes
 - Patient or bed partner notes limb movements during sleep.
 - Snoring
- ☐ Restless leg syndrome^[S] and **1 or more** of the following:^[N]
 - Excessive daytime sleepiness
 - Inadequate response to treatment
 - Observed apnea or choking episodes
 - Snoring

Alternatives

- For uncomplicated adult obstructive sleep apnea: home sleep study.(23)(24)(137) See Polysomnography (PSG), Home Sleep Study ^[AC] for further information.

Evidence Summary

Background

Sleep-disordered breathing is a common disorder affecting both children and adults; it is characterized by periods of breathing cessation (apnea) and reduced breathing (hypopnea). The most common form of sleep apnea, obstructive sleep apnea, is caused by the partial or complete collapse of the upper airway during sleep.(2) (**EG 2**) Obstructive sleep apnea is characterized by excessive daytime sleepiness, snoring, and witnessed apnea. Various scoring systems, such as the Epworth Sleepiness Scale and the Berlin Questionnaire, have been used clinically to screen for obstructive sleep apnea.(3)(4)(5) (**EG 2**)

The gold standard for diagnosis of obstructive sleep apnea is nocturnal (overnight) PSG in a sleep center laboratory that uses multiple channels to record a wide range of physiologic information, including electroencephalography, electrooculography, electromyography, electrocardiography, oronasal airflow, oxygen saturation, and esophageal pressure monitoring.(6)(7)(8) (**EG 2**) The apnea-hypopnea index is defined as the number of obstructive apneas and hypopneas per hour of recorded sleep; the respiratory disturbance index is the total number of breathing disturbances per hour of monitoring time (ie, the number of apneas, hypopneas, and respiratory event-related arousals, the last being characterized by increased respiratory efforts leading to arousals from sleep that do not fulfill the criteria for apnea or hypopnea). In adults, mild obstructive sleep apnea is defined as an apnea-hypopnea index or respiratory disturbance index

of 5 to less than 15, moderate as 15 to less than 30, and severe as 30 or greater. A sleep technician typically monitors the progress and quality of the sleep center study.(9)(10) **(EG 2)** The criteria for interpreting pediatric polysomnograms typically define mild obstructive sleep apnea as an apnea-hypopnea index of 1 to 5, moderate as 6 to 10, and severe as greater than 10.(7)(11)(12) **(EG 2)** A specialty society recommended using the pediatric scoring rules for children younger than 18 years; however, studies indicated that some of the adult scoring rules may be used in adolescents 13 to 18 years of age.(13)(14)(15)(16) **(EG 2)**

Criteria

The evidence for the clinical indications found in this guideline includes 101 published peer reviewed articles, 16 specialty society or other evidence-based guidelines, and 18 book sections.

For adults with obesity hypoventilation syndrome, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** A review article states that overnight oximetry and PSG are important diagnostic modalities for obesity hypoventilation syndrome because some patients only have nocturnal symptoms or manifestations; PSG can also detect coexisting obstructive and central sleep apnea.(20) **(EG 2)** A review article notes that the presentation of obesity hypoventilation syndrome more commonly includes dyspnea and low oxygen saturation during wakefulness as compared with that of obstructive sleep apnea.(21) **(EG 2)**

For adults with obstructive sleep apnea, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Overnight PSG is usually required for diagnosis; performance of overnight attended PSG in a sleep center remains the gold standard for diagnosis of obstructive sleep apnea.(2)(8)(9) **(EG 2)** Although overnight oximetry is sometimes used instead of PSG, the specificity of this test for moderately severe obstructive sleep apnea can be as low as 40%, requiring further testing in symptomatic patients with a negative result.(26) **(EG 2)** Sleep center PSG may be performed with CPAP titration as part of a split-night study for patients with suspected obstructive sleep apnea who are age 12 years or older.(27)(28)(29) **(EG 2)** However, a split-night study can only be performed if the diagnosis of moderate or severe obstructive sleep apnea can be made within the first 2 hours of recorded sleep, and at least 3 hours of CPAP titration is demonstrated, including the ability of CPAP to eliminate respiratory events during both rapid eye movement sleep and non-rapid eye movement sleep.(27) **(EG 2)** An observational study of 114 consecutive full-night polysomnograms suggested that data from the first 2 to 3 hours of testing may be sufficient to rule in obstructive sleep apnea at an apnea-hypoxia index threshold of only 5.(30) **(EG 2)** Specialty society guidelines note that follow-up PSG is appropriate in adult patients with moderate-to-severe obstructive sleep apnea who appear to have a good clinical response to an oral appliance or surgical intervention in order to document adequate therapeutic response; if symptoms return following an initial improvement with CPAP, surgery, or dental treatment; or after 10% or greater gain or loss of body weight in patients using CPAP to evaluate setting efficacy and ongoing need for therapy.(31)(32) **(EG 2)**

For central sleep apnea or sleep-related hypoventilation, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Central sleep apnea and central hypoventilation are diagnosed by full-night PSG.(13)(61) **(EG 2)** A review article notes that central sleep apnea is an independent risk factor for worsening heart failure and reduced survival in heart failure patients. Due to symptom overlap between central sleep apnea and heart failure, overnight sleep study is appropriate to diagnose central sleep apnea in these patients.(64) **(EG 2)** A specialty society practice parameter recommends PSG for suspected congenital central alveolar hypoventilation syndrome or suspected sleep-related hypoventilation due to neuromuscular or chest wall disorders.(57) **(EG 2)** Late-onset congenital central hypoventilation syndrome may present as hypoventilation, cyanosis, or seizure after anesthesia, sedation, or pulmonary infection.(65)(66) **(EG 2)** A retrospective review found that children with Prader-Willi syndrome experienced asymptomatic hypercapnic central sleep apnea and sleep-related hypoventilation more frequently than they experienced obstructive sleep apnea.(67) **(EG 2)** Review articles note that patients with Chiari malformation type 1 may present with sleep-disordered breathing caused by either central sleep apnea or obstructive sleep apnea.(68)(69) **(EG 2)** Children with undiagnosed congenital central alveolar hypoventilation syndrome may present with end organ damage due to hypoxemia or hypercarbia (eg, cor pulmonale, developmental delay, seizures) or unexplained laboratory abnormalities consistent with chronic hypercarbia or hypoxemia.(66) **(EG 2)**

For a child, infant, or adolescent with obstructive sleep apnea, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Pediatric obstructive sleep apnea affects between 1% and 6% of all school-age children.(87) **(EG 2)** A specialty society recommended using the pediatric scoring rules for children younger than 18 years; however, studies indicated that some of the adult scoring rules may be used in adolescents 13 to 18 years of age.(13)(14)(15)(16) **(EG 2)** Systematic reviews and meta-analyses of pediatric obstructive sleep apnea patients concluded that symptoms and signs (single or combined) and clinical scoring instruments that evaluated sleep-disordered breathing were unable to accurately predict a diagnosis of pediatric obstructive sleep apnea.(88)(89) **(EG 1)** Specialty society practice parameters discourage the use of nap, or abbreviated, PSG for evaluation. These parameters also discourage the routine use of PSG for the management of oxygen therapy in children being considered for treatment with supplemental oxygen.(57)(90) **(EG 2)** Specialty society guidelines recommend PSG, if available, prior to adenotonsillectomy in children being treated for obstructive sleep apnea for the purposes of diagnosis and preoperative risk assessment.(57)(85) **(EG 2)** Another specialty society guideline and a scientific statement recommend PSG prior to tonsillectomy in patients with obesity, craniofacial abnormalities, Down syndrome, mucopolysaccharidoses, neuromuscular disorders, or sickle cell disease,(91)(92) and in any patient when the need for tonsillectomy is not certain or the history and physical examination appear contradictory.(91) **(EG 2)** A retrospective chart review of interventions for obstructive sleep apnea in 126 neonates and infants age 0 to 12 months noted that supraglottoplasty was the most common surgical intervention in the first 3 months of life, and that the need for earlier surgical correction of craniofacial anomalies was identified by abnormal PSG.(93) **(EG 2)** A single-center retrospective study of PSG in 122 children with Down syndrome noted an overall prevalence of obstructive sleep apnea of 66.4%; in 52 children without suggestive clinical history, 53.8% had obstructive sleep

apnea documented on PSG.(94) **(EG 2)** A retrospective chart review of 97 children younger than 1 year who were referred to a sleep laboratory due to suspicion of obstructive sleep apnea reported that snoring and nocturnal desaturation were the 2 most common indications for referral, and that risk factors for obstructive sleep apnea included hypotonia, genetic conditions (eg, Down syndrome, Beckwith-Wiedemann syndrome, Marfan syndrome), craniofacial anomalies, neuromuscular disease, gastroesophageal reflux disease, adenotonsillar hypertrophy, and laryngomalacia.(95) **(EG 2)** A review article highlighted the lack of validated pediatric instruments for assessing excessive daytime sleepiness.(96) **(EG 2)** Another review article noted that obstructive sleep apnea may manifest in children as hyperactivity, emotional difficulties, decreased academic performance, and difficulty concentrating; by contrast, daytime sleepiness, morning headache, memory impairment, and daytime fatigue are more common symptoms in adults.(97) **(EG 2)** A joint specialty society guideline notes that PSG is appropriate in children with pulmonary hypertension who are at risk of sleep-disordered breathing and in children who are responding poorly to therapy for pulmonary artery hypertension.(98) **(EG 2)** Specialty society guidelines note that repeat PSG may identify children whose growth has changed their need for pressure therapy.(12)(57) **(EG 2)** For postoperative assessment after performance of surgery to treat sleep apnea in a child, a specialty society states that postoperative PSG is indicated in some children who undergo adenotonsillectomy for obstructive sleep apnea syndrome, as certain populations have been found to have incomplete resolution of disease; these higher-risk populations include children with craniofacial anomalies that obstruct the upper airway, certain neurologic disorders, and rapid maxillary expansion.(57) **(EG 2)** A secondary analysis of the data from a multicenter randomized controlled trial of adenotonsillectomy vs watchful waiting in 398 children with obstructive sleep apnea found that polysomnographic changes contributed only a limited role in accounting for outcomes and warned against the use of empiric apnea-hypopnea index thresholds to assess patient outcomes after treatment.(99) **(EG 2)** A review article notes that a minority of Chiari malformation patients who undergo foramen magnum decompression surgery may experience residual sleep-disordered breathing and that a postoperative polysomnogram no sooner than 6 weeks after surgery facilitates the identification of patients requiring additional interventions.(37) **(EG 2)** Performance of a split-night study may be indicated if the diagnosis of moderate or severe obstructive sleep apnea can be made within the first 2 hours of recorded sleep, and at least 3 hours of CPAP titration is demonstrated, including the ability of CPAP to eliminate respiratory events during both rapid eye movement sleep and non-rapid eye movement sleep.(27)(30) **(EG 2)** An evidence-based specialty society guideline is unable to recommend split-night CPAP titration for children younger than 12 years due to a lack of data.(29) **(EG 2)** A specialty society guideline on managing respiratory morbidity in infants, children, and adolescents born preterm (ie, less than 37 weeks' gestation) recommends PSG for patients who have failure to thrive, persistent need for supplemental oxygen at age 2 years, or ongoing snoring.(100) **(EG 2)**

For insomnia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** PSG is not indicated for the routine evaluation of transient insomnia, chronic insomnia, or insomnia associated with sleep disorders unless there is suspicion of sleep-disordered breathing or periodic limb movement disorder, a failure of insomnia treatments, or if precipitous arousals occur with injurious behavior.(119) **(EG 2)**

For narcolepsy, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** PSG is indicated prior to a multiple sleep latency test to rule out additional significant sleep disorder. PSG also confirms adequate sleep duration immediately prior to a multiple sleep latency test, which eliminates sleep deprivation as a possible cause of a false-positive multiple sleep latency test result.(123)(124) **(EG 2)** A retrospective chart review of 100 narcolepsy patients found that PSG helped to describe a high prevalence of sleep comorbidity in this population, including sleep-related breathing disorders in 24% of patients and sleep-related movement disorders in 55% of patients.(125) **(EG 2)** A specialty society practice parameter notes that patients who fail to respond to stimulant medication, experience increasing symptoms, or develop new sleep-related symptoms should be evaluated by PSG for comorbid sleep disorders.(126) **(EG 2)**

For parasomnia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** The primary role of PSG in parasomnias is to rule out specific conditions as the causes of nocturnal behaviors; video monitoring can be helpful. Rapid eye movement (REM) sleep behavior disorder is characterized by the acting out of dreams with violent or potentially injurious sleep-related behaviors (to the patient or others); it is typically diagnosed with PSG, which shows absence of muscle atonia during REM sleep.(2) **(EG 2)** Parasomnias may be diagnosed by history and clinical examination; however, a video polysomnogram may be required to further characterize atypical parasomnias, such as nocturnal epilepsy.(2)(130) **(EG 2)** A specialty society best-practice guideline includes abnormal PSG results as one potential criterion for a diagnosis of REM sleep behavior disorder.(131) **(EG 2)** The presence of REM sleep without atonia on polysomnogram is considered a biomarker for dementia with Lewy bodies, especially when accompanied by a history of REM sleep behavior disorder in the setting of dementia.(132) **(EG 2)**

For periodic limb movement disorder, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** PSG is recommended when there are complaints of repetitive limb movements during sleep, frequent awakenings, fragmented sleep, difficulty maintaining sleep, or excessive daytime sleepiness.(134) **(EG 2)** The addition of anterior tibialis surface electromyogram during PSG helps to differentiate limb movements due to this syndrome from those seen with respiratory events.(134)(135) **(EG 2)**

For restless leg syndrome, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** PSG is not routinely indicated for diagnosis or treatment; reliance on patient history or bed partner observations is usually sufficient for diagnosis and initiation of treatment. PSG may be indicated for restless leg syndrome if an additional sleep-related respiratory disorder is suspected or if symptoms do not respond to treatment.(2)(134)(136) **(EG 2)**

Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

Code of Federal Regulations (CFR): 42 CFR 419.22(138); 42 CFR 422.101(139)

Internet-Only Manual (IOM) Citations: CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 14 - Medical Devices(140); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 15 - Covered Medical and Other Health Services(141); CMS IOM Publication 100-02, Medicare Benefit Policy Manual, Chapter 16 - General Exclusions from Coverage(142)

Medicare Coverage Determinations: Medicare Coverage Database(143)

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Footnotes

[A] Hypoventilation syndromes include congenital central hypoventilation syndrome and obesity hypoventilation syndrome, as well as sleep hypoventilation observed in advanced stages of various obstructive lung diseases (eg, asthma, chronic obstructive pulmonary disease) and advanced stages of restrictive lung disease due to neuromuscular disease or lung parenchymal disease.(39) [A in Context Link 1, 2]

[B] Studies evaluating home polysomnography generally excluded patients with lung impairment that was classified by spirometry as moderate-to-severe category or worse, including patients with chronic obstructive pulmonary disease.(24) The threshold FEV₁ percent predicted for categorizing obstructive or restrictive lung disease as moderate-to-severe starts at a FEV₁ of less than 60% predicted (a category worse than the spirometry category of "moderate").(40) [B in Context Link 1, 2]

[C] Studies evaluating home polysomnography generally excluded patients with moderate-to-severe heart failure.(23) New York Heart Association functional class III patients are comfortable at rest, with less than ordinary activity causing symptoms. Class IV patients are unable to perform any physical activity and may be symptomatic at rest.(41) [C in Context Link 1, 2]

[D] A specialty society guideline recommends that sleep center polysomnography be performed when a home sleep study is negative despite excessive daytime sleepiness accompanied by additional suggestive findings (ie, habitual loud snoring, witnessed apneas or gasping or choking, diagnosed hypertension).(24) [D in Context Link 1, 2]

[E] A specialty society guideline recommends that sleep center polysomnography be performed when a home sleep study is inconclusive or technically inadequate.(24) [E in Context Link 1, 2]

[F] A split-night study includes a diagnostic polysomnography and CPAP initiation in the same night, if the individual demonstrates moderate-to-severe apnea (ie, apnea-hypopnea index of 15 or greater using adult criteria) over a minimum of 2 hours of polysomnography that night and the titration can last at least 3 hours during that study in individuals age 12 years or older.(24)(27)(29) [F in Context Link 1, 2]

[G] The Epworth Sleepiness Scale grades the chances of falling asleep (from 0 to 3) during 8 sedentary activities, such as watching television.(9) [G in Context Link 1, 2]

[H] Resistant hypertension is defined as persistently high clinic blood pressure despite patient adherence to a medication regimen consisting of 3 or more antihypertensive drugs with complementary mechanisms of action (a diuretic drug should be one component).(48) [H in Context Link 1, 2]

[I] Pulse oximetry may underestimate the degree of hypoxemia in patients with dark skin pigmentation, and the oximeter's ability to acquire an adequate signal may be impacted by cold body temperature, motion artifact, or presence of nail polish. Interpretation of the oximeter reading should take into account the clinical context (eg, level of anemia, altitude at which the reading is performed). Warming the patient, removing nail polish, and performing arterial blood gas analysis may be necessary to accurately determine oxygen status in some clinical situations.(52)(53)(54) [I in Context Link 1, 2, 3]

[J] A specialty society guideline notes that repeat sleep study may be appropriate in patients found to have CPAP device-generated event data that are unexplained (eg, treatment-emergent central sleep apnea) or who develop cardiovascular disease or experience a change in cardiovascular comorbidity status for which repeat polysomnography may result in a change of treatment plan (eg, newly discovered arrhythmia or significant conduction defect, stroke or transient ischemic attack while prescribed CPAP therapy).(32) [J in Context Link 1]

[K] Low arterial oxygen saturation (SpO₂) suggestive of nocturnal hypoventilation is defined as nocturnal SpO₂ below 90% for more than 5 minutes accompanied by SpO₂ nadir of at least 85%, or SpO₂ below 90% for more than 30% of total sleep time.(79)(80) [K in Context Link 1]

[L] Elevated PaCO₂ is defined as values above the upper limits of normal (ie, greater than 45 mm Hg (6.0 kPa)).(81) Erythrocytosis can occur in response to hypoxemia and may be suggested by a hematocrit above 48% (0.48) in adult females, above 52% (0.52) in adult males, above 50% (0.50) in children, and above 65% (0.65) in neonates in the first week of life.(82)(83)(84) [L in Context Link 1]

[M] Split-night studies include a diagnostic PSG study plus a CPAP titration study; these may occur when the laboratory threshold for moderate or severe obstructive sleep apnea has been demonstrated for a sufficient time period at the start of the study. Moderate or severe obstructive sleep apnea in adolescents using pediatric criteria includes apnea-hypopnea index of 6 or greater. The sparsity of data in children younger than 12 years, the success of adenotonsillectomy for many pediatric obstructive sleep apnea patients, and the need to desensitize many pediatric patients to positive airway pressure and the mask interface account for the less frequent use of split-night study in pediatric patients.(29)(115)(116) [M in Context Link 1, 2]

[N] A specialty society guideline notes that repeat sleep study may be appropriate in patients found to have CPAP device-generated event data that are unexplained (eg, treatment-emergent central sleep apnea).(32) In addition, the need to repeat polysomnography in children may occur when academic difficulties, behavioral problems, cognitive deficits, reduced growth rate or growth failure (ie, weight or height below the fifth percentile for age and sex), pulmonary hypertension, or systemic hypertension raises suspicion of persistent obstructive sleep apnea.(12) [N in Context Link 1]

[O] Insomnia is defined as difficulty initiating or maintaining sleep, waking up too early, inadequate sleep time, or sleep that is chronically nonrestorative or poor in quality.(2) [O in Context Link 1, 2]

[P] Sleep-disordered breathing is most commonly obstructive apnea, but it may also include hypoventilation manifested by an increase in PaCO₂ during sleep of more than 10 mm Hg (1.3 kPa) above the value while awake or significant oxygen desaturation that is not explained by obstructive apneas or hypopneas.(120) [P in Context Link 1]

[Q] Parasomnias are abnormal behaviors associated with sleep that are classified by the sleep state in which they occur (ie, non-rapid eye movement sleep or rapid eye movement sleep) and caused by activation or inhibition of functional groups of neurons.(2) [Q in Context Link 1, 2]

[R] Periodic limb movement disorder is characterized by periodic episodes of repetitive and highly stereotyped limb movements that occur during sleep; it is associated with insomnia or daytime tiredness.(2) [R in Context Link 1, 2]

[S] Restless leg syndrome is a disorder characterized by an irresistible urge to move the legs, which may be accompanied by disagreeable leg sensations; the urge usually starts or worsens with inactivity or rest, and may be most pronounced during, or isolated to, the evening or night. It is a clinical diagnosis; other medical or behavioral disorders that may present in a similar manner should be ruled out.(2) [S in Context Link 1, 2]

Codes

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